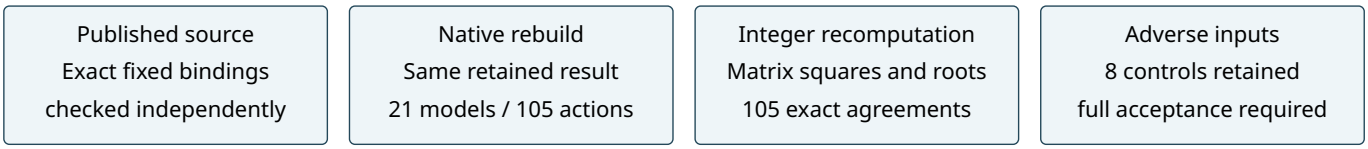
Independent retained Kummer matrix review

A separate offline native build reproduced the producer result exactly. Independent Python integer arithmetic recomputed all 105 matrix-square identities and normalized roots across the 21 retained models, with identical ordered results. Eight adverse copied-input controls were checked; producer bindings remained unchanged.



Source fidelity. Published checker, Cargo files and acceptance predicate were decoded and matched to commit 80b98db381 before review execution. The native checker uses an explicit 21-prime list, exact five-action membership, canonical four-by-four matrices, prime p<=2^32, widened products and four-term sums, and ordered scalar comparison. Its largest matrix-square sum is below 2^66. Root normalization uses modular exponentiation only after prime and nonzero-denominator checks. The independent calculation uses arbitrary integers and modular inverse. No point or trajectory enumeration was performed.

Acceptance boundary. Six structural or algebraic corruptions fail with native exit1: wrong matrix shape, noncanonical entry, changed coefficient, duplicate model, missing model and reordered scalar certificates. Removing one nonessential historical pin and its index entry can yield native accepted=true, but the separate twelve-pin acceptance predicate rejects it. Replacing the supplied walk file with unrelated text also yields native accepted=true and passes that predicate: the program records its new hash but does not enforce the externally approved hash. The full review therefore requires exact externally frozen source/input bindings in addition to native success and the inventory predicate. This control is retained rather than described as a native rejection.

Historical source correspondence. The recorded source versions differ in cached versus uncached theta inversion. The retained extracted loader and linear sections are byte-identical; the independent review also compares those bytes to the bound full sources. The checker result's prose about source agreement is a fixed description, not a dynamic proof for arbitrary supplied walk files. For the frozen files, exact hashes and the separate section comparison supply the source statement. This does not attest an executed collector binary or its runtime input provenance.

Scope and access. The reviewer prepared the private inputs and prospective plan, and had prior source familiarity; this is an independent implementation review, not blind selection. Recompute code was prepared before publication, but all review execution and producer source inspection followed publication. The producer packet and failures are unchanged. The scientific campaign and 1,260-target obligation are unchanged; no new timing, cohort or cost result is generated.